

PROTEIN MODELING

COMPUTATIONAL BIOPHYSICS WORKSHOP 2026

PRESENTED BY

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AMINO ACID SEQUENCES

- ❑ Information regarding the proteins function (catalytic activity, specific recognition sites, etc.), and properties (molecular weight, isoelectric point (pI), extinction coefficient).
- ❑ The proteins evolutionary origin.
- ❑ Information regarding the type of its 3D structure (folding type).
- ❑ Proteins with the same function taken from closely related organisms have highly similar amino acid sequences.

COMMON CHANGES OF A PROTEIN SEQUENCE

- ❑ Replacement of one amino acid by another
- ❑ Insertions and deletion of single amino acids and larger blocks
- ❑ Large rearrangements of the gene (less common)

TYPES OF SEQUENCE COMPARISONS

- ❑ Compare sequence to database - goal: find related sequences (search)
- ❑ Compare sequence to sequence - goal: find matching parts
(ALIGNMENT)
- ❑ Multiple sequence alignment and phylogenetic analysis (evolutionary trees)

2.4 Comparing sequences

The comparison of biological sequences is one of the most important operations in computational biology.

Fundamental assumption:: High sequence similarity implies similar structure and/or function

In an *alignment* of two sequences: place one sequence above the other one such that similar or identical characters are in the same column and non-identical/non-similar characters are either placed in the same column as a mismatch or opposite a gap in the other sequence.

Example:

Two strings:	→	Alignment:
IMISSMISSISSIPPI		I-MISSMISSISIPPI-
MYMISSISAHIPPIE		MYMISS-ISAH-IPPIE

OTHER GOALS OF SEQUENCE COMPARISONS

- ❑ Finding and using a consensus sequence – a highly invariant stretch of amino acids
- ❑ Examples
 - Enzyme active sites
 - Co-factor binding sites
 - Other functionally important sites
 - Sites of post-translational modifications: glycosylation, cleavage of signal sequences etc.

SEQUENCE ALIGNMENT

- ❑ There are many ways of aligning two sequences.
- ❑ An alignment produced by a program is one of many possible and we need **criteria to judge the quality of a given alignment.**

ALIGNMENT ACCURACY

- ❑ The best alignment is the one that has the maximum number of identical residues aligned against each other - **% identity**.

- ❑ Example:

Sequence 1 Sequence 2

CPKICIGGWFAAY CSGICKKAWFV – Y

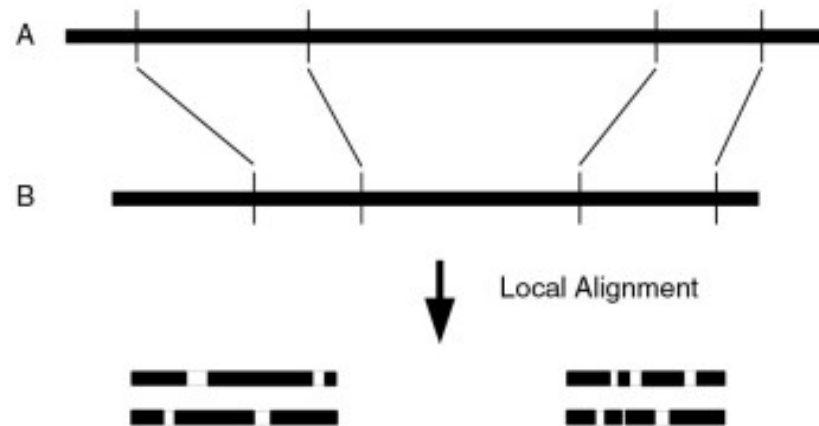
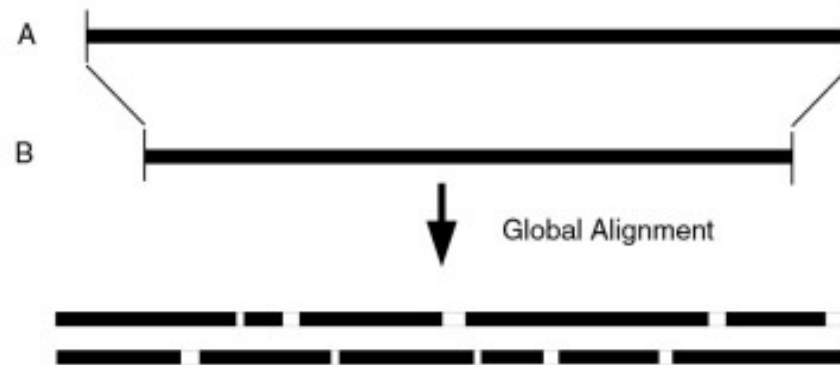
- ❑ Alignment pattern : **C – – IC – – – WF – – Y**

- ❑ Identity = $6/13 = 46\%$

- ❑ Score (s) = matches – mismatches = $6 - 7 = -1$

TYPES OF ALIGNMENT

- ❑ **Global alignment:** align two sequences from beginning to end, insisting that all sequence positions must be matched. Used in the alignment of sequences known to be related.
- ❑ **Local alignment:** find the best region of similarity between two sequences without insisting that the entire sequences match (a result will be several alignments with close or different scores). Used in database searching and in alignment of distantly related sequences with several regions of homology.



4.1 Multiple sequence alignment (MSA)

A multiple sequence alignment (MSA) is simply an alignment of more than two sequences, like this:

```
MRP2_HUMAN      TSNRWLAIRLELVGNLTVFFSALMMVIY--RDTLSGDTVGFVLSNALNITQTLNWLVRMT
Q9UQ99_HUMAN     VANRWLAVRLECVGNCIVLFAALFAVIS--RHSLSAGLVGLSVSYSLQVTTYLNWLVRMS
ABCC8_HUMAN      AANRWLEVRMEYIGACVVLIAAVTSISNSLHRELSAGLVGLGLTYALMVSNYLNWMVRNL
Q96J65_HUMAN     CALRWFALRMDVLMNLTFTVALLVTLS--FSSISTSSKGLSLSYIIQLSGLLQVCVRTG
Q96JA6_HUMAN     SSTRWMALRLEIMTNLVTLAVALFVAFG--ISSTPYSFKVMVAVNIVLQLASSFQATARIG
MRP5_HUMAN       CAMRWLAVRDLISIALITTTGLMIVLM--HGQIPPAYAGLAISYAVQLTGLFQFTVRLA
MRP4_HUMAN       TTSRWFVRDLAICAMFVIIVAFGSLIL--AKTLDAGQVGLALSVALTLMGMFQWCVRQS
O75555_HUMAN     TTSRWFVRDLAICAMFVIIVAFGSLIL--AKTLDAGQVGLALSVALTLMGMFQWCVRQS
CFTR_HUMAN       STLRFQMRIEMIFVIFFIIVTFISILT---TGEGERVGIILTLAMNIMSTLQWAVNSS
```

(A small section of a multiple alignment of the human CFTR protein and eight homologous proteins.)

Multiple sequence alignment is applied to a set of sequences that are assumed to be related and the goal is to detect homologous residues and to place them in the same column of the multiple alignment.

Multiple alignments are more suitable than pairwise alignments to address evolutionary questions, as the chance of random similarities occurring decreases, as the number of aligned sequences grows.

Quote (Arthur Lesk): One or two homologous sequences whisper ... a full multiple sequence alignment shouts out loud...

FUNCTIONAL INFORMATION FROM MULTIPLE SEQUENCE ALIGNMENT

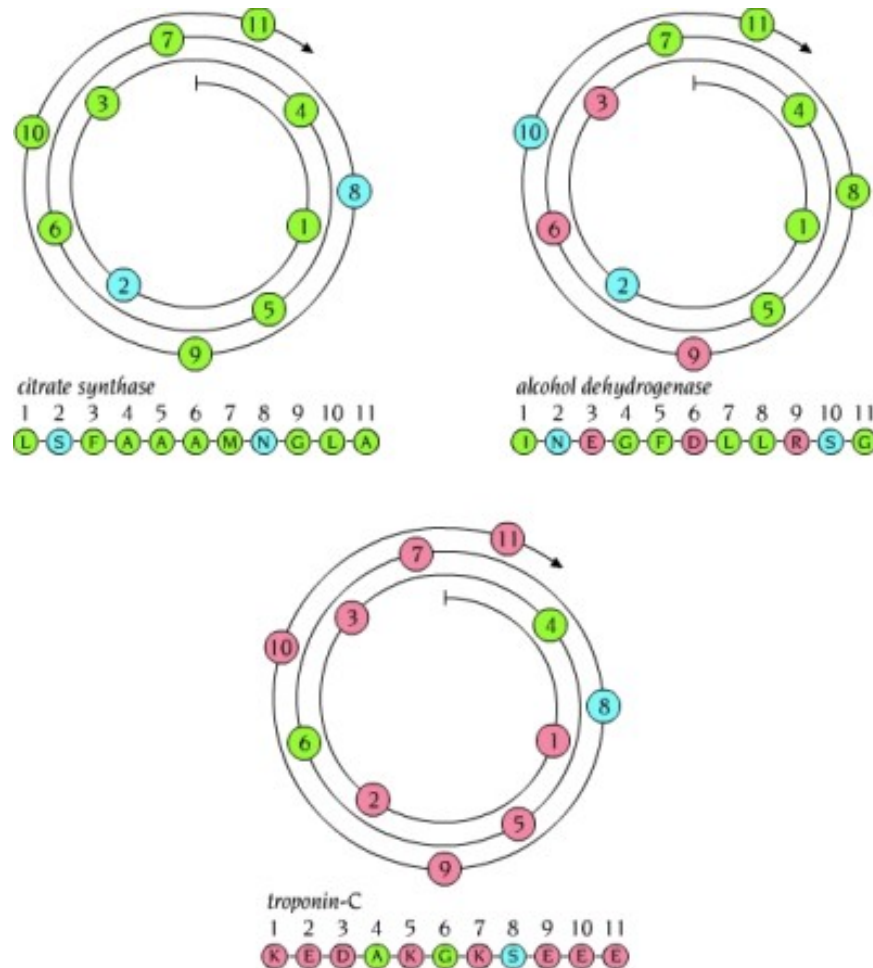
- ❑ A multiple sequence alignment allows us to extract information which is difficult to extract from a single sequence or from an alignment of only two sequences.
- ❑ When making multiple sequence alignment, if possible, choose sequences from organisms belonging to different kingdoms of life (bacteria, archaea, eukaryotes) and group the sequences accordingly.
- ❑ This will allow you to deduce the conservation pattern within each group and across all kingdoms.

Barley	- -MAAMATALS T S L P H L P F R R L P S H P Y A A L S L A P R G F R R R E A P A R L A A V A S A S E Y L D S T N G S 39
Pea	H N F S L T H T - P H T T A S F N L Q L R F H S L L P F S - F T S Q P F L S L H S T P P F K R T Y P K L R - - A Q S E N G S 37
Tobacco	H N F C S T S L P Q T S L S N S Q S S T F F T Y L K P C P I L S S T F L R P K R L K P R L R I S A T A T - - I D S P M G S 39
Synchr	-----M T L T P F I P -----
Rcsp	-----
Barley	A A P A P T S F A P S G Q Q K T R E Y F P L A A Y V G Q D A I K T S L L L G A I D R - E Y G G I A I S Q R G T A K T V 119
Pea	A V L Q A S E E K - L D A S H T R N Q V F P L A A Y I R Q D A I K T A L L L G A T D F - N I G G I A I S Q R G T A K T I 116
Tobacco	A V A V V E P E K Q P E K I S F R Q V F P L A A Y I G Q D A I K T A L L L G A I D R - E I G G I A I C Q R G T A K T L 119
Synchr	-----L N E R I T A I V D Q E A I K L A L L L G A I D F - G L G G I V I A Q R G T A K S V S I
Rcsp	-----M D H E R L S A L A V L T V D A A V G L W L R S R A P I R L A S S
Barley	H A R G L H A M L P P I E V V V S I A N A D P N I P E E W E D H L A D Q V Q Y D A D G N V K C E I V K A P F V Q I P L G 110
Pea	H A R G M H A I L P P I E V V Q S I A N A D P S C P E E W E D G L Y R R V E Y D S D G N V K T H I I K S P F V Q I P L G 117
Tobacco	H A R G L H A I L P P I E V V V S M A N A D P N C P D E W E D G L A D R A E Y G S D G N I K T Q I V K S P F V Q I P L G 110
Synchr	H A B A I H T L L P P I E I I K G N V Q C D P K N R G S W D D D T L E K P A D V P L D Q L E T Q V I P A R E I Q I P L G 112
Rcsp	P T D T L A K L P F M A L R -----R L E P N S S
Barley	V T E D R L I S S V D V E S V K S T T V F Q P E L L A E A H R S V L Y V D E I N L L D E G I S H L L L N Y L T E G V R 24
Pea	V T E D R L I S S V D V E S V K T T T V F Q P E L L A E A H R S V L Y V D E I N L L D E G I S H L L L N Y L T E G V R 236
Tobacco	V T E D R L I S S V D V E S V K S T T V F Q P E L L A E A H R S V L Y V D E I N L L D E G I S H L L L N Y L T E G V R 24
Synchr	V T E D R L I S S V D V E S V K Q S A V E Q P E L L A Q A H R S V L Y I D E L N L L D Q Q I A N Q L L T V L T E G K N 17
Rcsp	V D G G A L Y G L D W A E T L N G S K P V L K G E L L D R P S V F I L P M A R -----C T A K L G 10
Barley	I V E R E G I S P R H P C K P L L I A T Y N P E S S V R E H L L D R I A I N L S A D L P L S F D D R Y A A V H I A T Q F 302
Pea	I V E R E G I S P R H P C R P L L I A T Y N P D E S S V R E H L L D R I A I N L S A D L P M S F E N R Y E A V G I A T E F 299
Tobacco	I V E R E G I S P R H P C K P L L I A T Y N P E S S A V R E H L L D R I A I N L S A D L P M S F D D R Y A A V D I A T R F 302
Synchr	G I E R E G H S F Q H P C O P L S I A T Y N P E S P L R H L L D R I A I A L S A D G I L G L D Q R Y A A Y D G V L A Y 304
Rcsp	A R L A Q A L D L Q H A L I A L D E A A E D E - A L P H A V A D H L S L F V D L S E V R S I D S -----I 151
Barley	Q E S S K D V F K M V E E T E V A K T Q I L A R E Y L K D V A I S T S Q L K Y L V M E A I N S G C Q G H R A S L Y A A 340
Pea	Q D N C S Q V F K M V D E D T D N A K T Q I L A R E Y L K D V T I S R E Q L K Y L V I E A L N S G V Q G H R A S L Y A A 340
Tobacco	Q E C S N E V F K M V D E S T D F A K T Q I L A R E Y L K D V T I S R D Q L K Y L V M E A I N S G C Q G H R A S L Y A A 340
Synchr	A D S P I S F I D Q T D A E L D O L E T T I L A R E M L K E V S L T P E Q V S Y L V E A I N S G L Q G H R G L F A M 325
Rcsp	-----P S L L P E T A Q I E R A R E L L P O V O M P A S R V S E I M E R C R Q L S I S S L A P M L A L 200
Barley	R V A K C L A M E G R E K V F A E D L K K A V E L V I L P R S I I S D N P Q S Q Q N G P P P P P P P P Q N G D N A E 434
Pea	R V A K C L A L E G R E K V F V D L K K A V E L V I L P R S I I T D T P P E Q Q N G - P P P P P P P P - Q N Q S N E 419
Tobacco	R V A K C L A X I D G R E K V G V D L K K A V E L V I L P R S T I V E N P P D Q Q N G Q P P P P P P P - Q N Q D S S E 420
Synchr	R V A K A I A A L D G R S D V Q A D D L R Q A V E L V I V E R S V L M D N R P - - P P E Q A P P P P P P - Q N Q D E K 320
Rcsp	T A A R I T A L S G R T R M A E D V L H A A L L T L A N A L P -----L Q E A P P P P P P P P P P P P M L 220
Barley	D Q D E K E E D E K D E E E K D D D E E N E K D D Q I P E E F I P D A E S G L V D E K L L P F A Q Q A Q R R K G K A 48
Pea	E Q N E E E E Q E - - E E E - E D D N D E E N E Q D Q D O L P E E F I P D A E S G L V D E K L L P F A Q Q A Q R R R G K A 477
Tobacco	E Q N E E E E K E - - E E D Q E D E K D R E N E Q D Q Q V P D E F I P D A E S G L V D E K L L P F A Q Q A Q R R K G K A 480
Synchr	D E Q E D Q Q D D - - - K E D D K D N P E A E Q D P P S I P E E F I E D P E G V S L D P S V L Y R A G M A Q - K Q G E S 410
Rcsp	G E N Q Q D E Q D - - - - - Q I D P L D G I P E I V V E A V R A H L P D N I L Q T L N H G S R L R A A S 30
Barley	G R - - A K N V I P S E D N G R Y I K P H L P C G P - V R L A V D A T L R A X A P Y Q K L R R - E K S L D K T R K M P 540
Pea	G R - - A K N V I P S E D N G R Y I K P H L P C G P - V K R L A V D A T L R A X A P Y Q K L R R - E K D T E N R K M V 534
Tobacco	G R - - A K K V I P S E D N G R Y I K P H L P C G P - V K R L A V D A T L R A X A P Y Q K L R R - A K D I Q K T R K M V 539
Synchr	G - - S R S V I P S D D R G R Y L E R I L P K E K - M R R I A V D A T L R A X S P Y Q S R R - L R H P D - - R O M I V 444
Rcsp	G G G G A G Q Q I G N R R G R L P R R G S L E D D A K I D L V A L L R S A A R M G L E S R Q A P A S T E R V L L 340
Barley	E C T D M R A K R M A R K A G A L V I F V V D A S S M A L N R N Q N A K G A L K L L A E S Y T S R D Q V A I I P F R G 400
Pea	E C T D M R A K R M A R K A G A L V I F V V D A S S M A L N R N Q N A K G A L K L L A E S Y T S R D Q V S I I P F R G 398
Tobacco	E C T D M R A K R M A R K A G A L V I F V V D A S S M A L N R N Q N A K G A L K L L A E S Y T S R D Q V C I I P F R G 400
Synchr	E Q S D I N G K L V R K A G A L I V F L V D A S S M A L N R N Q A K G A R V M Q L L T E A Y E N R D Q V S L I P F Q S 525
Rcsp	E S S D I H I K R R K E N S D R V L I F A V D A S S A A V A L S E A R G A V E L L L O R A Y A A R D H S L I T E R G 420

STRUCTURAL INFORMATION CAN BE DERIVED FROM MULTIPLE SEQUENCE ALIGNMENT

- ❑ Positions of insertions and deletions suggest regions of surface loops in the 3D structure.
- ❑ Conserved Gly and Pro suggest a β -turn.
- ❑ Hydrophobic residues conserved at $i, i + 2, i + 4$ etc separated by hydrophilic residues suggest a surface β -strand.
- ❑ A short run of hydrophobic residues (4 *aa*) may suggest a buried β -strand, longer stretches (20 *aa*) may suggest a membrane spanning helix.
- ❑ Pairs of conserved hydrophobic *aa* separated by pairs of hydrophilic residues suggest an α -helix with one face packed against the protein core (visualised with a helical wheel)

INTO THE LOCATION OF THE HELIX IN A STRUCTURE



4.1.2 Conservation of structural elements

For example, here is an alignment of N-acetylglucosamine-binding proteins:

```
AATAHAQR G EQGSNME PN NL CC SQYGY GMGGDY GK G .. QNGA YT
VAATNAQT G KQNDGMI PH NL CC SQFGY GLGRDY GT G .. QSGA CS
VGLVSAQR G SQGGGGT PA LW CC SIWGW GDSEPY GRT .. ENK . CWS
AATAQAQR G EQGSNME PN NL CC SQYGY GMGGDY GK G .. QNGA WT
AATAQAQR G EQGSNME PN NL CC SQYGY GMGGDY GK G .. QNGA WT
.....QR G EQSGME PN NL CC SQYGY GMGGDY GK G .. QNGA WT
SETVKSQN G ..... AP NL CC SQFGY GSTDAY GT G .. RSGP CRS
RGSAE . . Q G RQAGDAL PG GL CC SSYGW GTTVDY GIG .. QSQ . DG
AGPAAAQN G ..... QP NF CC SKFGY GTTDAY GDG .. QSGP CRS
AGPAAAQN G ..... QP NV CC SKFGY GTTDEY GDG .. QSGP CRS
RGSAE . . Q G RQAGDAL PG GL CC SSYGW GTTADY GDG .. QSQ . DG
RGSAE . . Q G RQAGDAL PG GL CC SFYGW GTTVDY GDG .. QSQ . DG
TGVAIAEQ G RQAGGKL PN NL CC SQWGW GSTDEY SPD HNK QSN . K .
.....EQ G RQAGGKL PN NL CC SQYGW GSDDY SPS KNK QSN . K .
```

The alignment shows 8 conserved cysteins. These form 4 bridges, which are essential to stabilize the protein:



A new sequence should only be added to the family if it has this pattern of conserved cysteins, which can be seen using a multiple sequence alignment.

SEARCHING DATABASES

- ❑ Homology search - Used to search for related sequences in sequence databases and allows a search with long sequences, programs like *FastA* and *Blast* are used.
- ❑ Pattern Searches - used to find short sequence patterns in a single sequence, in a group of sequences or in a database.
- ❑ Pattern searches only decides whether there is an exact match or not (subsequently no gaps or substitution matrices used).
- ❑ Useful when there is no information on a particular protein.

STEPS IN SEQUENCES ANALYSIS

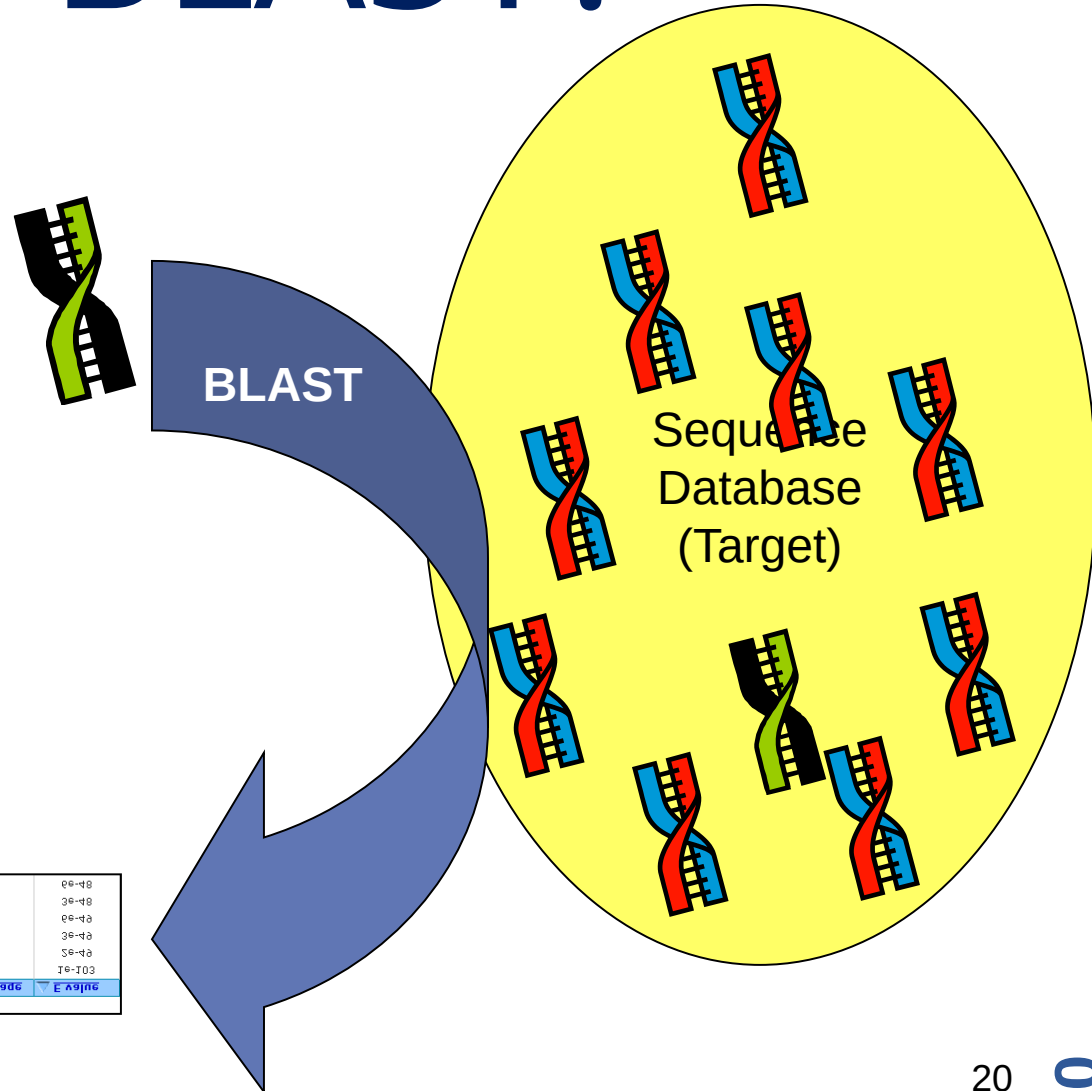
1. Find the sequence, e.g. in SwissProt.
2. Search databases for homologues using *BLAST* or *FASTA* (local alignment of short stretches). If convinced by the hits, go to step 4.
3. If the BLAST search was not 100% convincing, also search for patterns, motifs.
Compare with known function, if known, etc.
4. Found a sufficient number of homologues? Make a multiple sequence alignment
5. Search PDB. Any homologues with a known 3D structure ? Make a homology-based model - useful for a deeper understanding of function, design of mutations, etc.

BLAST

- ❑ BLAST (Basic Local Alignment Search Tool) is a set of similarity search programs designed to explore all of the available sequence databases regardless of whether the query is protein or DNA.
- ❑ Programs within BLAST:
 - *blastp* - Compares an amino acid query sequence against a protein sequence database.
 - *blastn* - Compares a nucleotide query sequence against a nucleotide sequence database.
 - *blastx* - Compares a nucleotide query sequence translated in all reading frames against a protein sequence database.
- ❑ This option can be used to find potential translation products of an unknown nucleotide sequence.

WHAT IS BLAST?

Query tool to retrieve homologous genes from a database



Putative Function

55 010708207	геномное [модифицированное СС-метил] >дР[ЕВМ-4210]	733	783	100%	98-98
CV0088837	qla [микроорганизмический СС-метил]	734	784	100%	98-98
16 4894847	геномное [модифицированное СС-метил] >дР[ЕВМ-4210]	733	783	98%	98-98
16 3378887	геномное [модифицированное СС-метил] >дР[ЕВМ-4210]	733	783	98%	98-98
16 007884527	геномное [модифицированное СС-метил] >дР[ЕВМ-4210]	733	783	100%	98-98
16 4400487	геномное [модифицированное СС-метил] >дР[ЕВМ-4210]	733	783	100%	98-98
Accession	Description	Max score	Total score	Query coverage	E value

Значения показаны в значащих единицах:

DIFFERENT FORMS OF BLAST

- *blastp*
- *blastn*
- *blastx*
- *tblastn*
- *Tblastx*
- *BLAST2*
- Biology Workbench (<http://workbench.sdsc.edu/>) is a good source of tools

STARTS WITH A QUERY SEQUENCE IN FASTA FORMAT

- ❑ Amino acid sequence of a protein, in FASTA format:

```
>ribosomal protein L7/L12 [Thiomicrospira crunogena XCL-2]
MAITKDDILEAVANMSVMEVVELVEAMEEKFGVSAAAVAVAGPAGDAGAA
GEEQTEFDVVLTGAGDNKVAAIKAVRGATGLGLKEAKSAVESAPFTLKEG
VSKEEAETLANELKEAGIEVEVK
```

- ❑ Nucleotide sequence of a gene, in FASTA format:

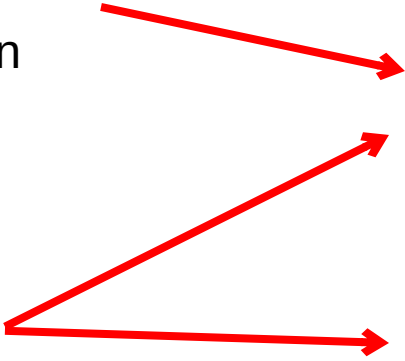
```
>gi|118139508:333094-333465 Thiomicrospira crunogena XCL-2
ATGGCAATTACAAAAGACGATATTTTAGAAGCAGTTGCTAACATGTCAGTAATGGAAGTTGTTG
AACTTGTTGAAGCAATGGAAGAGAAGTTTGGTGTCTTCTGCAGCAGCAGTTGCGGTTGCAGGTCC
TGCAGGTGATGCTGGCGCTGCTGGTGAAGAACAAACAGAGTTTGACGTTGTCTTGACTGGTGCT
GGTGACAACAAAGTTGCAGCAATCAAAGCCGTTTCGTGGCGCAACTGGTCTTGGGCTTAAAGAAG
CGAAAAGTGCAGTTGAAAGTGCACCATTTACGCTTAAAGAGGGTGTTTCTAAAGAAGAAGCAGA
AACTCTTGCAAATGAGCTTAAAGAAGCAGGTATTGAAGTCGAAGTTAAATAA
```

KEY ASPECTS OF FASTA FORMAT

"description line" (not read as sequence data)

- Begins with >
- Ends with a hard return

Sequence data
(amino acid in this case)



```
ribosomal proteinL7/L12 <  
MAITKDDILEAVANMSVMEVVVELVEA  
MEEKFGVSAAAVAVAGPAGDAGAA  
GEEQTEFDVVLTGAGDNKVAAIKAVR  
GATGLGLKEAKSAVESAPFTLKEG  
VSKEEAETLANELKEAGIEVEVK
```

NCBI BLAST INTERFACE (BLASTP: PROTEINS)

 **BLAST** *Basic Local Alignment Search Tool*

[Home](#) [Recent Results](#) [Saved Strategies](#) [Help](#)

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Enter Query Sequence

BLASTP programs search protein databases using a protein query. [more...](#) [Reset page](#) [Bookmark](#)

Enter accession number, gi, or FASTA sequence [Clear](#) Query subrange [Clear](#)

From

To

Or, upload file

Job Title

☐ Align two or more sequences [Help](#)

Choose Search Set

Database [Help](#)

Organism ☐ Exclude [+](#)

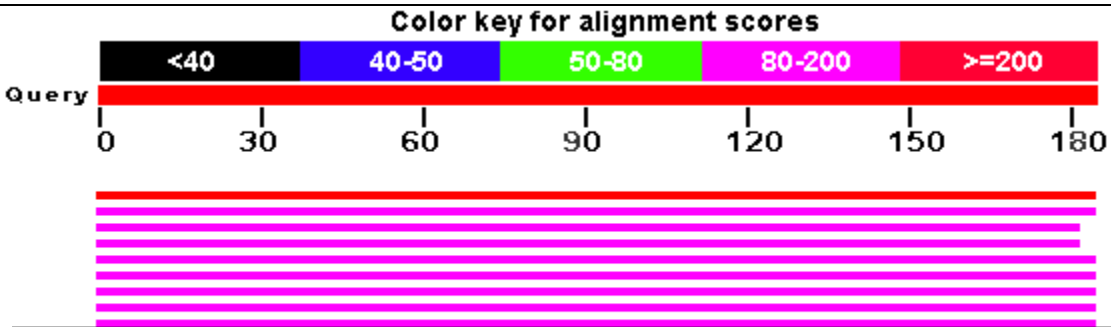
Optional Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown. [Help](#)

Exclude ☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Optional Entrez Query

Optional Enter an Entrez query to limit search [Help](#)

POTENTIAL HOMOLOGS RETRIEVED FROM DATABASE



Sequences producing significant alignments:

Accession	Description	Max score	Total score	Query coverage	E value
NP_440048.1	potential FMN-protein [Synechocystis sp. PCC 6803] >sp P727	379	379	100%	1e-103
YP_001864295.1	flavin reductase domain-containing protein [Nostoc punctiform	199	199	100%	2e-49
YP_321888.1	flavin reductase-like, FMN-binding [Anabaena variabilis ATCC	198	198	98%	3e-49
NP_488484.1	flavoprotein [Nostoc sp. PCC 7120] >sp Q8YNW7.1 DFA4_ANA	197	197	98%	6e-49
CAO89562.1	dfa4 [Microcystis aeruginosa PCC 7806]	194	194	100%	3e-48
ZP_01630850.1	flavoprotein [Nodularia spumigena CCY9414] >gb EAW44518.	193	193	100%	6e-48

```
>ref|YP\_002482587.1| G flavin reductase domain protein FMN-binding [Cyanotheca sp. PCC 7425]
gb|ACL44226.1| G flavin reductase domain protein FMN-binding [Cyanotheca sp. PCC 7425]
Length=585

GENE ID: 7287783 Cyan7425_1859 | flavin reductase domain protein FMN-binding [Cyanotheca sp. PCC 7425]

Score = 176 bits (446), Expect = 1e-42, Method: Compositional matrix adjust.
Identities = 95/196 (48%), Positives = 125/196 (63%), Gaps = 16/196 (8%)

Query 1   SGANFARQLRTHKRQRIARQATTETQADRTQQAVGRIIGSIGVTTQTTRH----- 52
          +G++FA+ L+ K+QR RQ+ E Q+DRT+QAVGRIIGS+ V+T + H
Sbjct 393  AGSDFAQVLKKAKKQKRSRQSIQVQSDRTEQAVGRIIGSLCVLTAKQQQTHPHPEVEEP 452

Query 53   -----QGILTSWVSQASFTPPGIMLAIPGEFDAYGLAQNKAFVLNLLQEGRSVRRHFDH 107
          +L SWVSQASF PPG+ +A+ E A GL AFVLN+L+EG ++RRHF
Sbjct 453  QLEVPTAMLVSWVSQASFNPGLTIALAKE-RAEGLDHSGDAFVLNVLKEGMNLRHFSK 511

Query 108  QPLPKGDGNPFSRLEHYSTQNGCLILAEALAYLECLVQSWSNIGDHVLYATVQAGQVLO 167
          P G++ P+ L +NGC +L + LAYLEC VQS GDH L+YATV G+VLQ
Sbjct 512  SFAP--GEDRFAGLNIIQAENGCPVLQDCLAYLECTVQSRMECDGHVLIYATVNMNGKVLQ 569

Query 168  PNGITAIRHRKSGGQY 183
          P G TA++HRKSG QY
Sbjct 570  PTGTTAVQHRKSGNQY 585
```

OVERVIEW OF BLAST

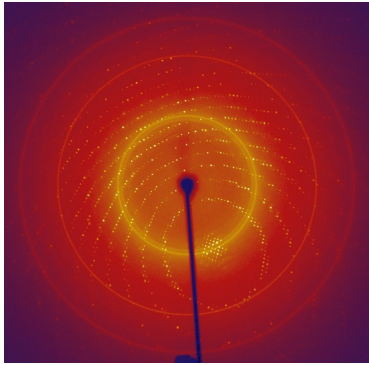
1. Segment the query sequence
2. Use the query sequence segments to scan the database
3. Extend the segments to verify the matching sequences (hits) from the database
4. Create a list of hits, with best matches first

E-VALUE

- Number of distinct alignments with scores greater than or equal to a given value expected to occur in a search against a database of known size, based solely on chance, not homology.
- Large E-values suggest that the query sequence and retrieved sequence similarities are due to chance
- Small E-values suggest that the sequence similarities are due to shared ancestry (or potentially convergent evolution)

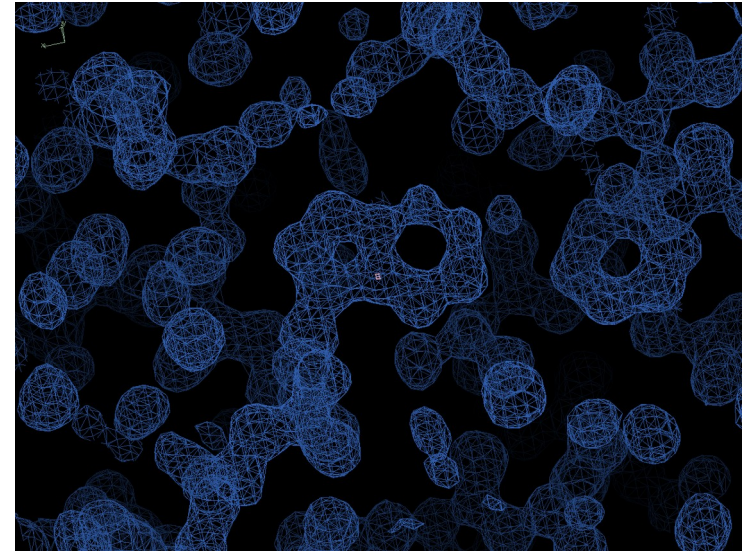
Protein Structure

A LOOK AT X-RAY DATA



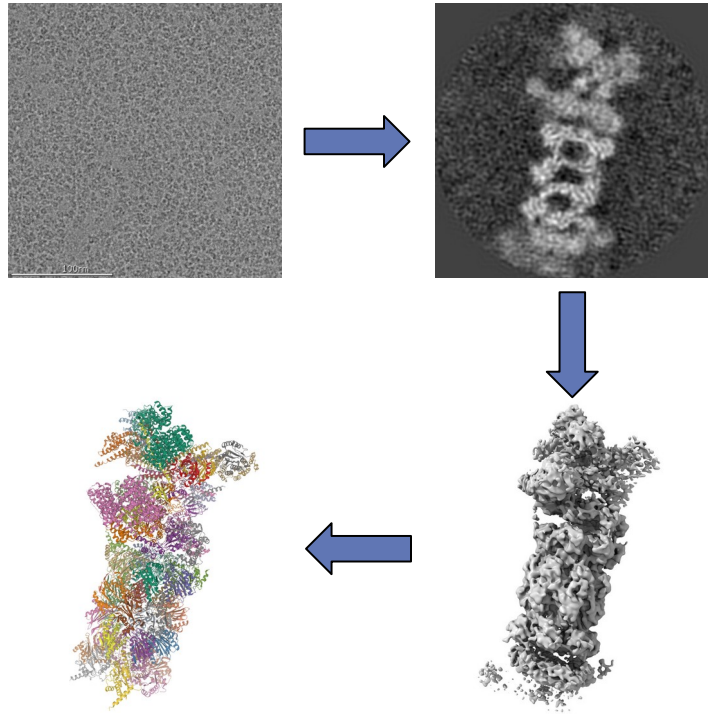
From the diffraction pattern, a map can be calculated.

This indicates the location of electrons (therefore atoms) in the crystal
Hence- 'electron density map'

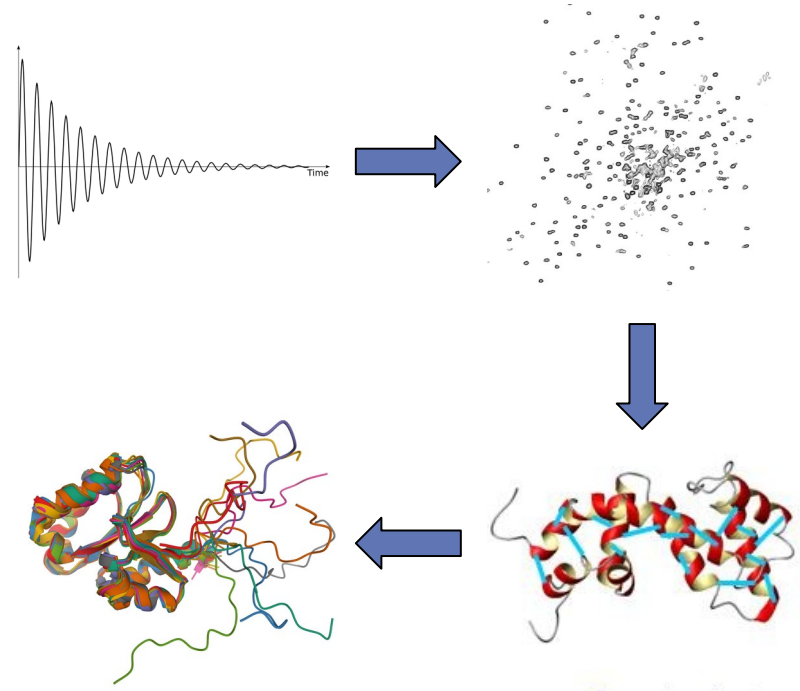


Model is built into this map in an iterative process

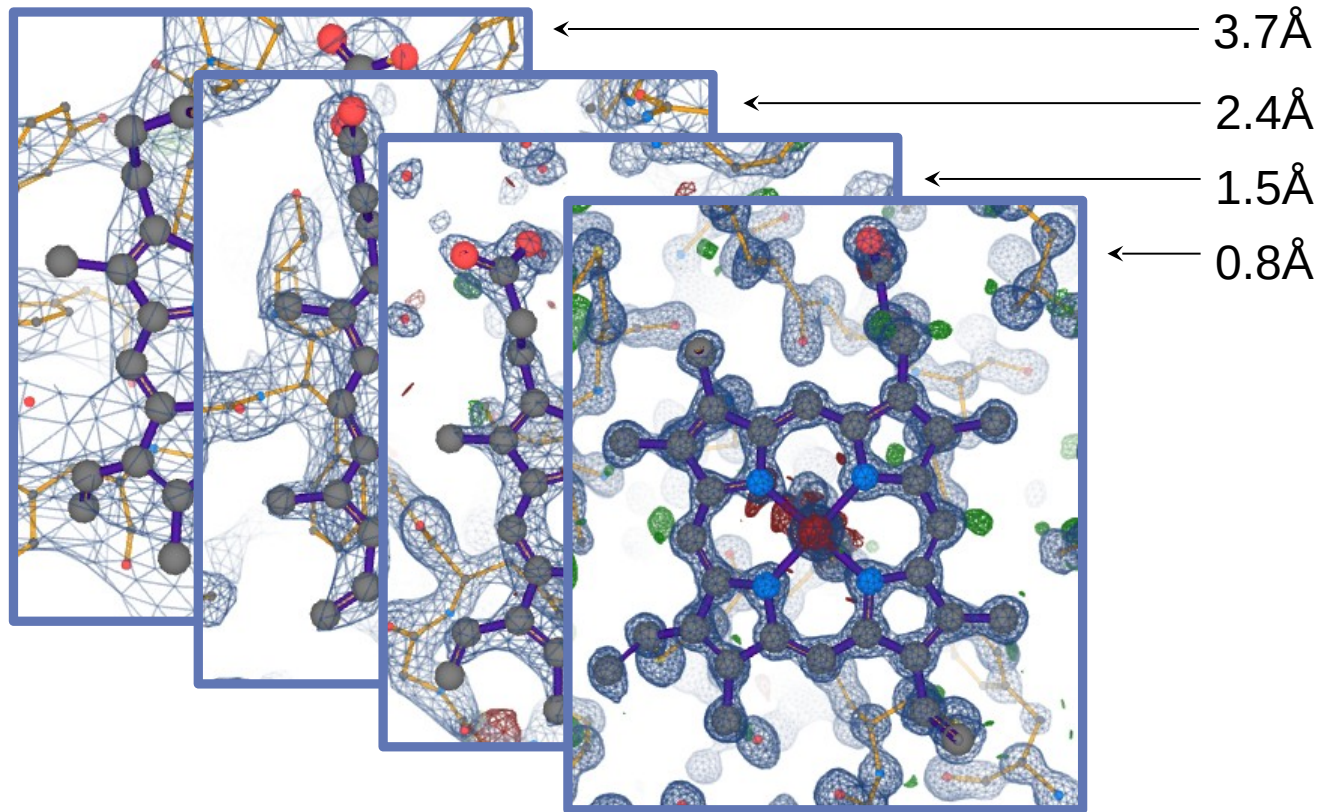
CRYO - ELECTRON MICROSCOPY (CRYOEM)



NUCLEAR MAGNETIC RESONANCE (NMR)



RESOLUTION INDICATES PRECISION WITH WHICH ATOMS CAN BE PLACED

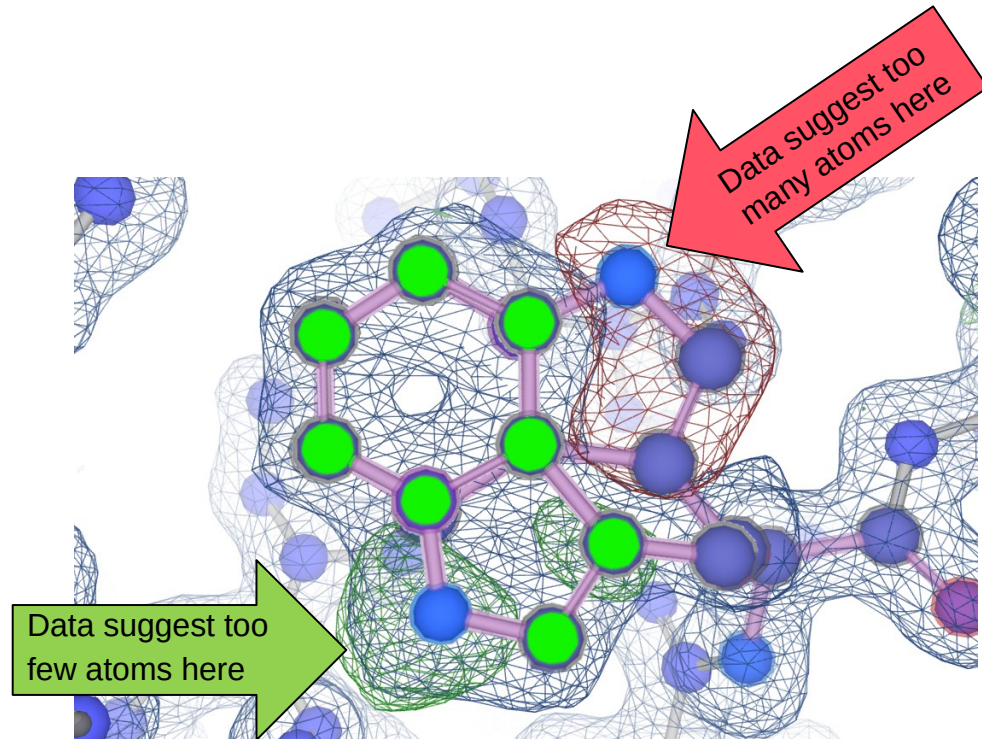


THE PDB FILE IS ONLY AN INTERPRETATION OF THE DATA

- Difference density helps identify where the model does not match the data:

Areas where the model has too many atoms for the data

Areas where the model has too few atoms for the data



The PDB file format

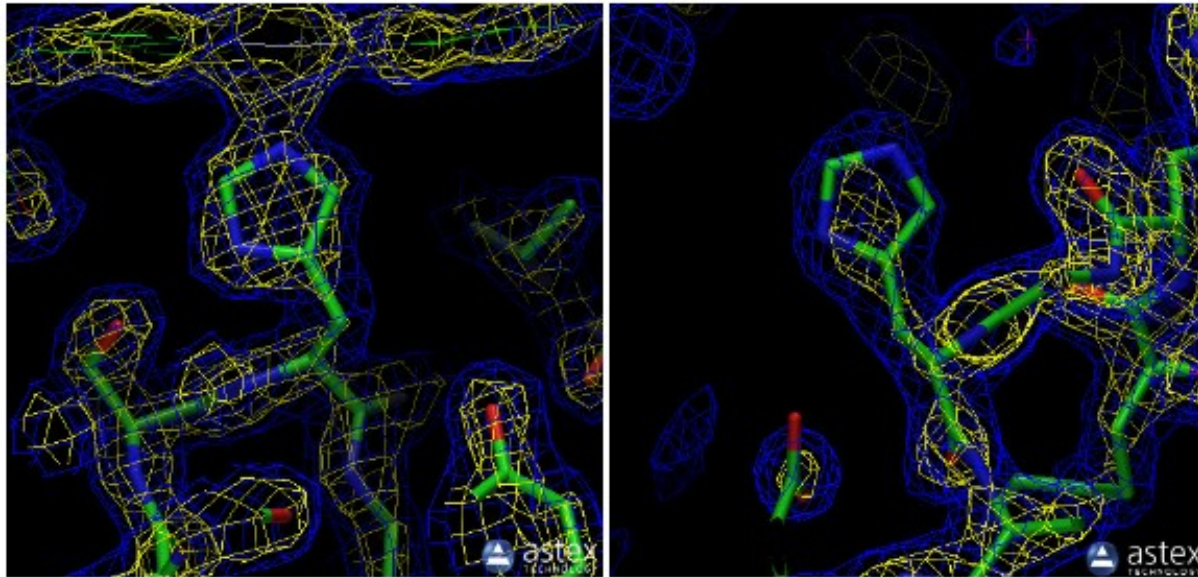
Just a text file! Fixed column width - Card legacy

Chain name	ATOM	2365	O	GLU	S	271	-11.042	-31.638	22.562	1.00	13.19	O
	ATOM	2366	CB	GLU	S	271	-9.351	-31.199	25.481	1.00	12.72	C
	ATOM	2367	CG	GLU	S	271	-10.019	-32.565	25.731	1.00	14.90	C
	ATOM	2368	CD	GLU	S	271	-10.069	-32.942	27.205	1.00	15.79	C
	ATOM	2369	OE1	GLU	S	271	-10.068	-34.150	27.487	1.00	21.99	O
	ATOM	2370	OE2	GLU	S	271	-10.101	-32.059	28.084	1.00	19.89	O
	ATOM	2371	N	ALA	S	272	-11.559	-29.817	23.813	1.00	11.35	N
	ATOM	2372	CA	ALA	S	272	-12.918	-29.744	23.288	1.00	10.72	C
	ATOM	2373	C	ALA	S	272	-12.958	-29.381	21.778	1.00	10.89	C
	ATOM	2374	O	ALA	S	272	-13.709	-29.992	21.005	1.00	10.98	O
Residue name	ATOM	2375	CB	ALA	S	272	-13.814	-28.804	24.129	1.00	11.26	C
	ATOM	2376	N	ALA	S	273	-12.102	-28.459	21.336	1.00	9.29	N
	ATOM	2377	CA	ALA	S	273	-12.087	-28.103	19.894	1.00	10.77	C
	ATOM	2378	C	ALA	S	273	-11.709	-29.274	18.936	1.00	10.55	C
	ATOM	2379	O	ALA	S	273	-12.179	-29.288	17.806	1.00	11.31	O
	ATOM	2380	CB	ALA	S	273	-11.222	-26.891	19.632	1.00	9.88	C
	ATOM	2381	N	ALA	S	274	-10.961	-30.295	19.389	1.00	12.12	N
Residue number	ATOM	2382	CA	ALA	S	274	-10.493	-31.383	18.449	1.00	14.23	C
	ATOM	2383	C	ALA	S	274	-11.100	-32.890	18.262	1.00	14.94	C
	ATOM	2384	O	ALA	S	274	-10.339	-33.762	18.453	1.00	13.37	O
	ATOM	2385	CB	ALA	S	274	-8.960	-31.501	18.670	1.00	13.94	C
	ATOM	2386	N	GLN	S	275	-12.304	-33.243	17.708	1.00	19.11	N
	ATOM	2387	CA	GLN	S	275	-12.815	-34.714	17.812	1.00	16.43	C
	ATOM	2388	C	GLN	S	275	-13.255	-35.682	16.572	1.00	17.45	C
	ATOM	2389	O	GLN	S	275	-13.460	-36.964	16.661	1.00	4.38	O
	ATOM	2390	CB	GLN	S	275	-13.871	-34.737	18.905	1.00	19.77	C
Atom name	ATOM	2391	CG	GLN	S	275	-14.310	-36.127	19.356	1.00	23.54	C
	HETATM	2396	ZN	ZN	S	278	-11.252	-10.370	14.483	1.00	28.39	ZN
	HETATM	2397	ZN	ZN	S	279	-10.199	-35.599	11.656	1.00	18.26	ZN
	HETATM	2398	ZN	ZN	S	280	16.091	-23.317	24.137	1.00	37.16	ZN
	HETATM	2399	ZN	ZN	S	281	2.562	-32.376	26.687	1.00	26.51	ZN

Coordinates of
atom in space (Å)

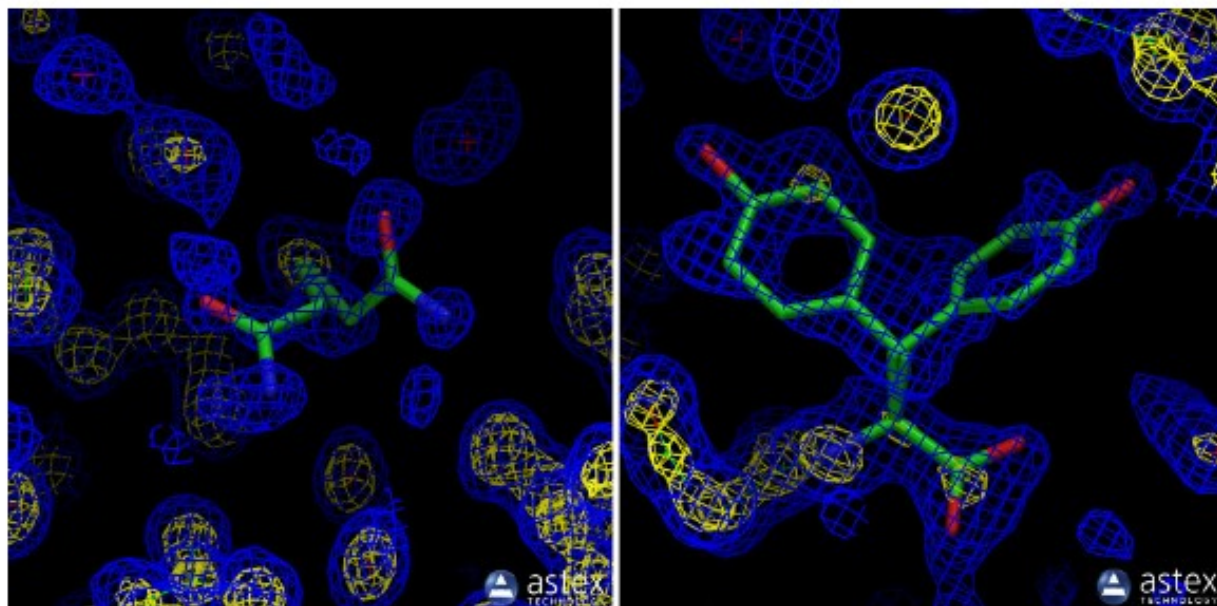
Occupancy

“Temperature”
value /B-factor



The example shown is from a myoglobin structure solved at 2.0 Å resolution (PDB entry 1mbi). Two histidine amino acids are shown. On the left is HIS93, which coordinates with the iron atom and thus, is held firmly in place. It has B-values in the range of 15-20—notice how the contours nicely surround the whole amino acid, revealing a sharp electron density. On the right is HIS81, which is exposed on the surface of the protein and has higher B-values in the range of 22-74. Also note how the contours enclose a smaller space, showing a smaller region with high electron density for this amino acid because the overall electron density is weakly smeared in the space around the contours.

<https://pdb101.rcsb.org/learn/guide-to-understanding-pdb-data/dealing-with-coordinates>

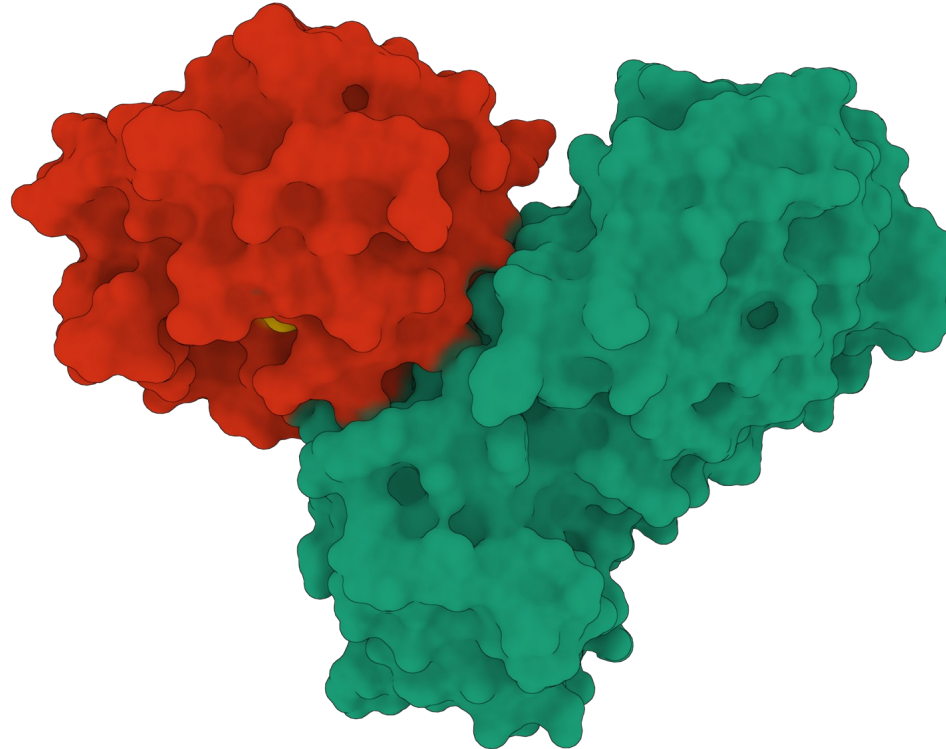


Alternate Conformations in Myoglobin: The two images shown are taken from the high-resolution structure of myoglobin in entry 1a6m: glutamine 8 is on the left, and tyrosine 151 on the right. In both cases, the depositors interpreted the experimental data as showing two conformations of the amino acid, with occupancies of 0.57 and 0.43 for the glutamine, and 0.5 for each of the tyrosine conformations. The blue contours surround the regions with high electron density, and the atomic model is shown in sticks.

<https://pdb101.rcsb.org/learn/guide-to-understanding-pdb-data/dealing-with-coordinates>

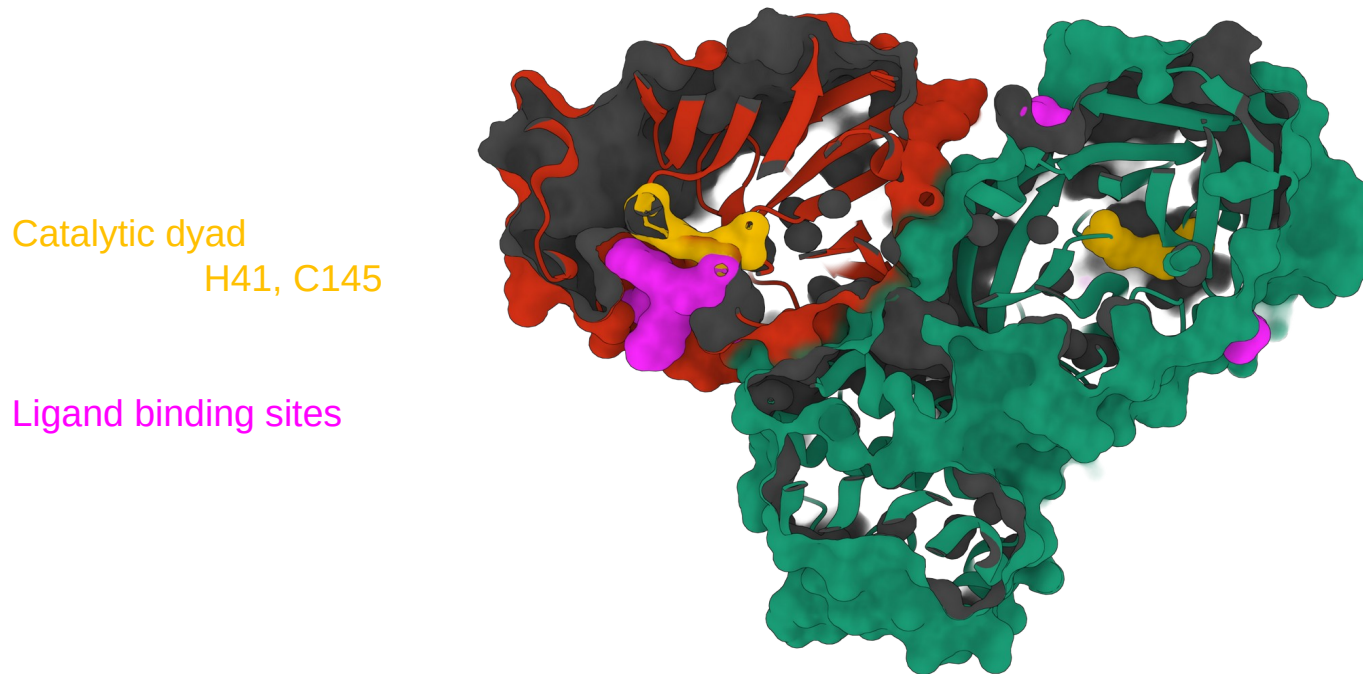
SARS-CoV-2 main protease

Homodimer



PDB 6LZE

SARS-CoV-2 main protease



PDB 6LZE

Such value added annotations gives scientific meaning to 3D coordinates

Structure Prediction and Modelling

STRUCTURE PREDICTION: FOUR MAIN PROBLEM AREAS

- ❑ Given a sequence with unknown structure, predict its structure

- ❑ Secondary structure prediction

Predict regions of helices and strands

- ❑ Homology modeling

Predict structure from known structures of one or more related proteins

- ❑ Fold recognition

Given a library of structures, determine which one (if any) is the fold of the given sequence

- ❑ Prediction of novel folds: A-priori and knowledge-based methods

STRUCTURE PREDICTION OF NOVEL FOLDS: TWO APPROACHES

❑ A priori:

Most approaches aim to reproduce inter-atomic interactions by defining an energy function and trying to find global minimum.

❑ Problem

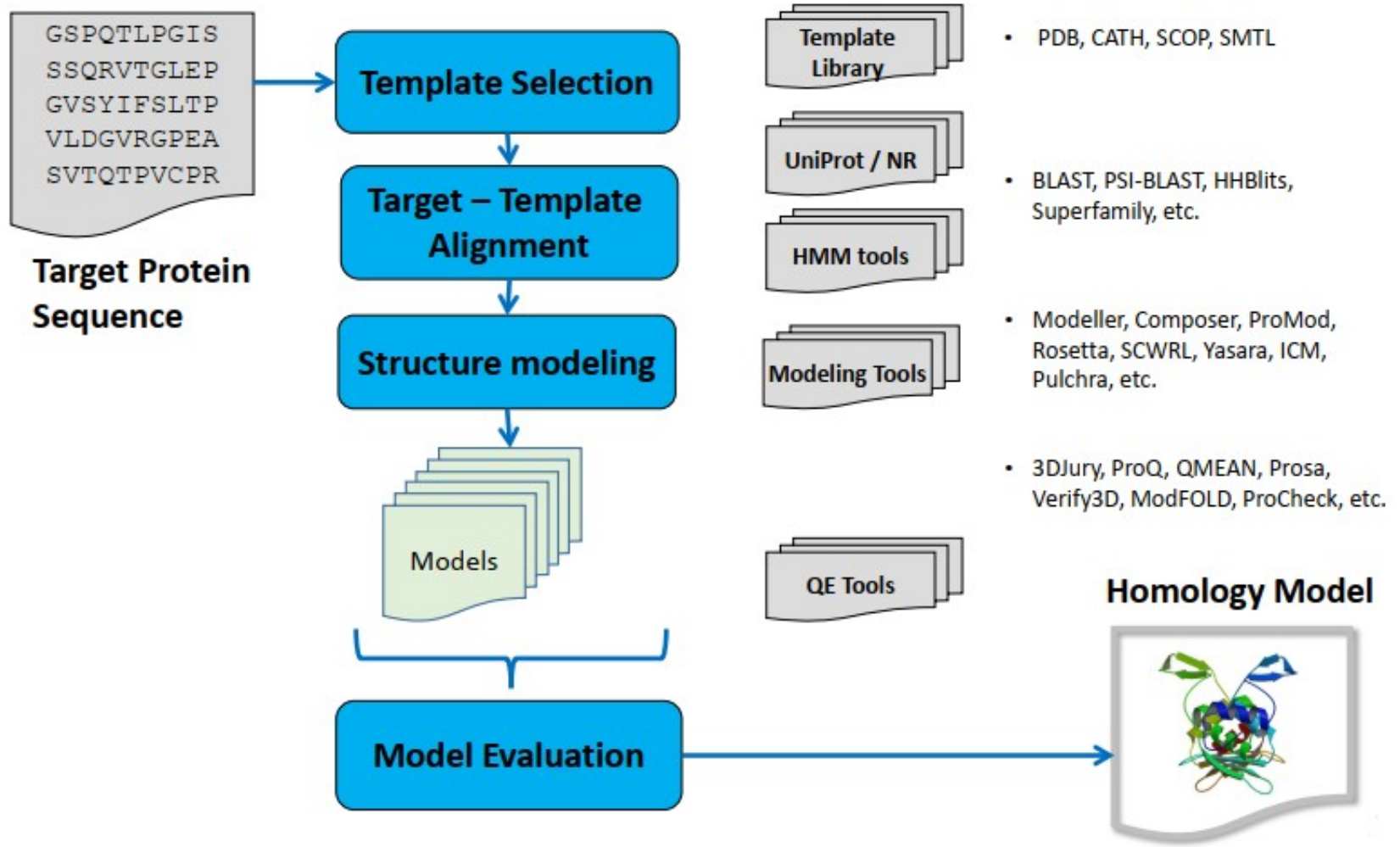
- I. Inadequacy of the energy function
- II. Search algorithms get stuck in local minima

❑ Knowledge - based: Find similarities to known structures or substructures

HOMOLOGY MODELLING

- ❑ Define the model of an unknown structure by making minimal changes to a relative with known structure
- ❑ Align amino acid sequences of target and one or more known structures
 - Insertions and deletions should be in loop regions
- ❑ Determine main chain segments to represent the regions containing insertions and deletions and stitch these into the known structure
 - Replace the side chains of the residues that have been mutated
- ❑ Examine the model (by hand and computationally) to detect collisions between atoms
- ❑ Refine the model by limited energy minimization

Homology Modelling Workflow



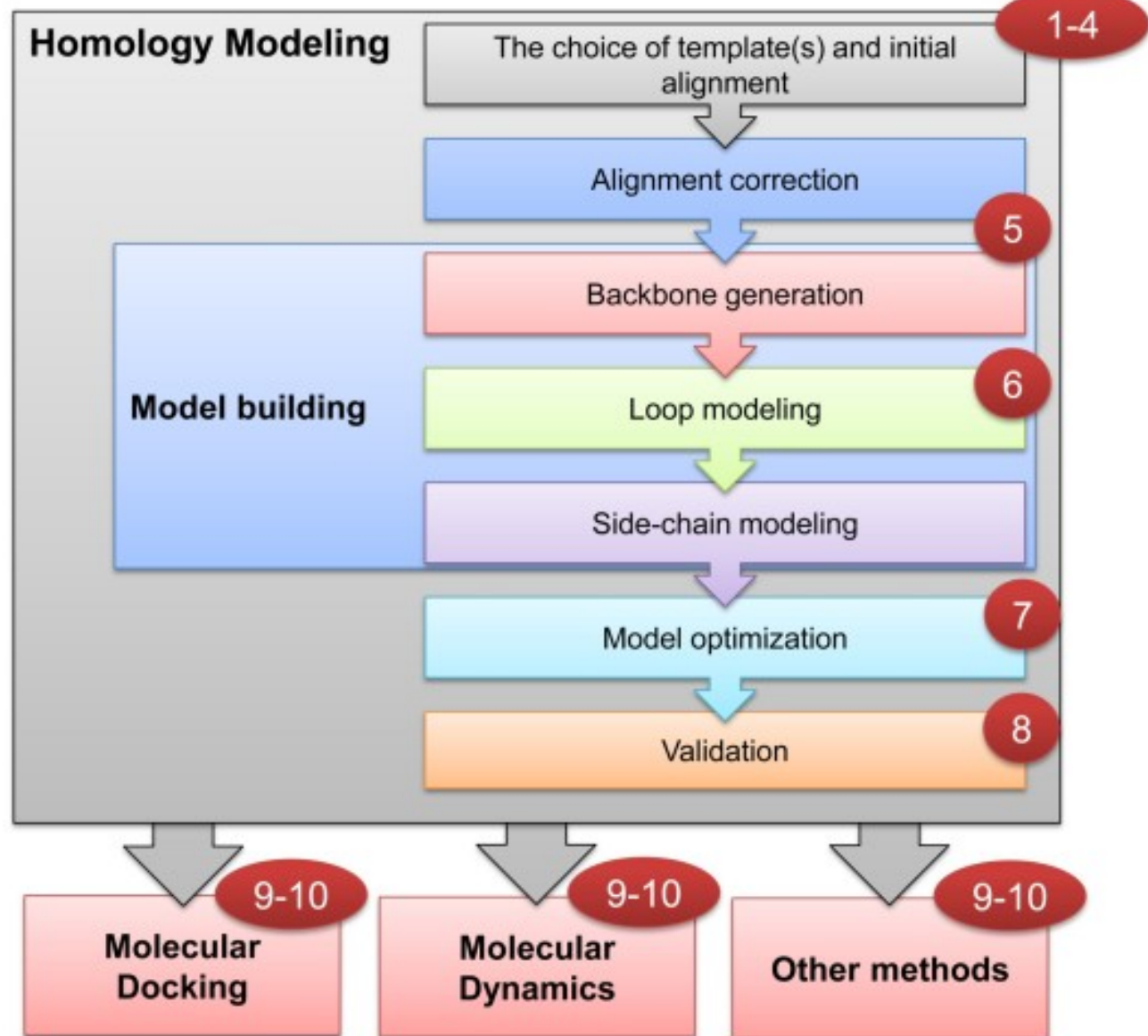
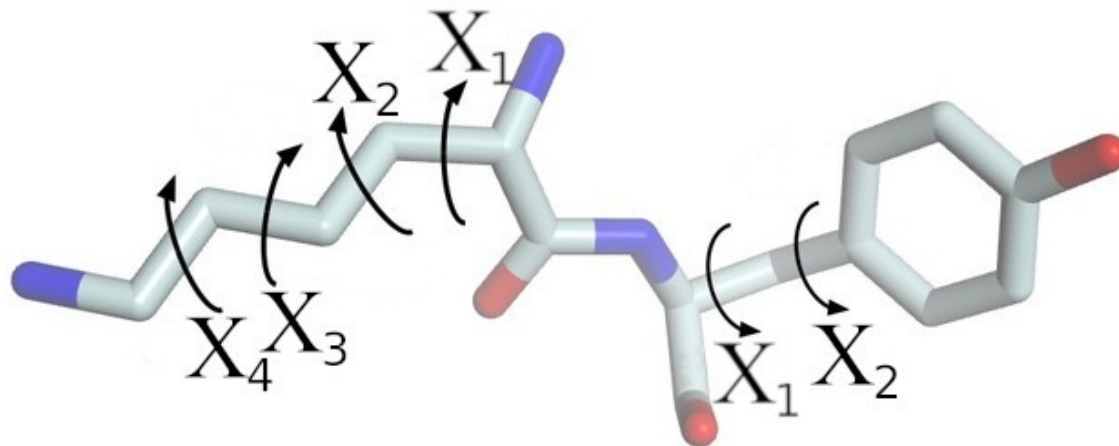
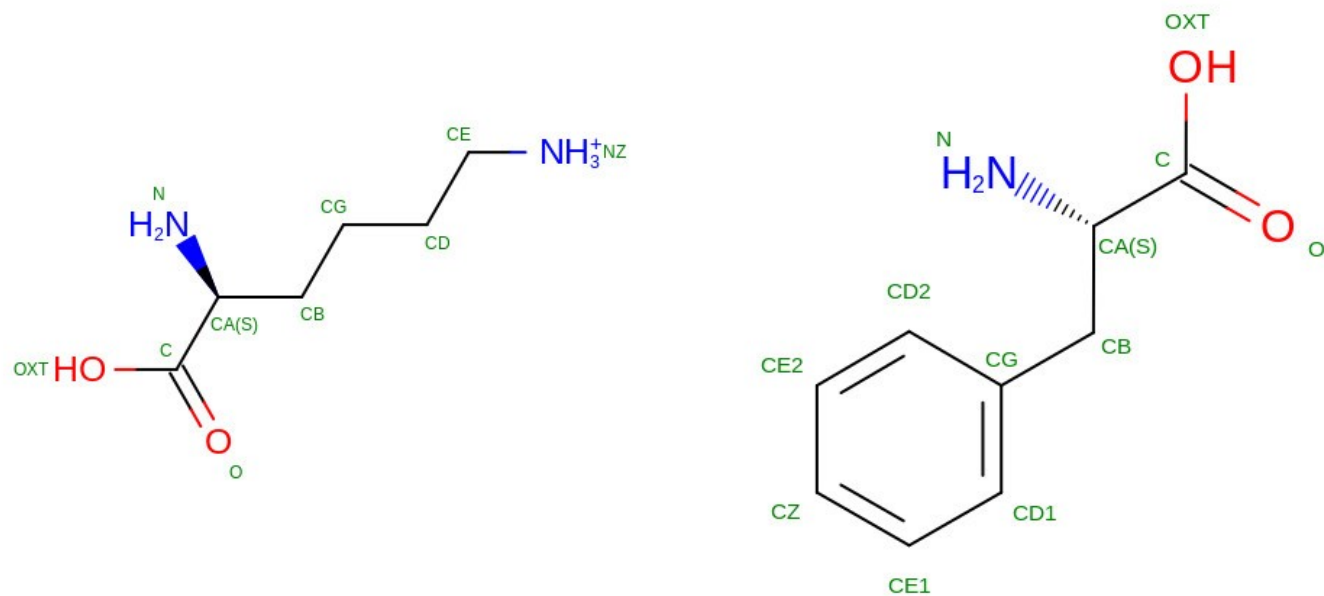
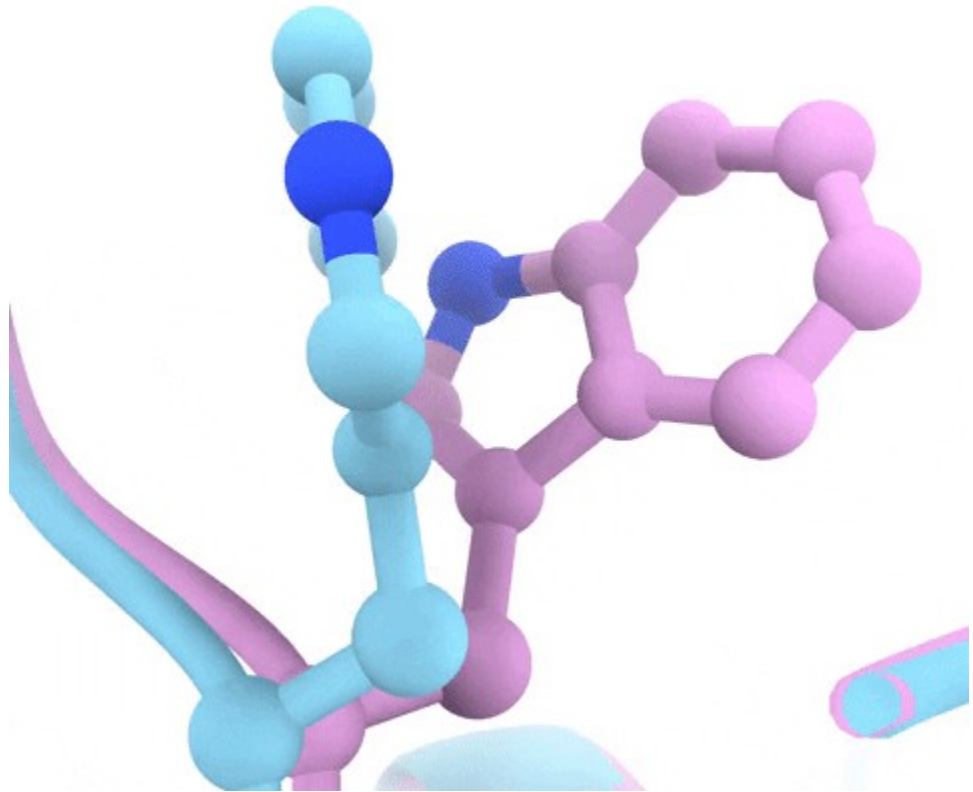
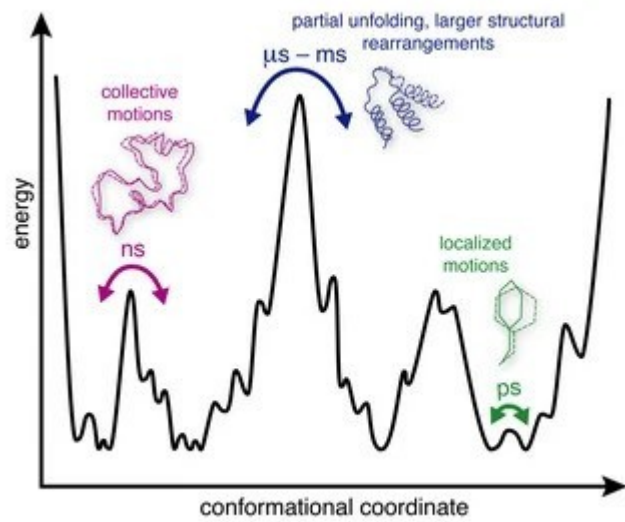


Fig 1. Homology-modeling workflow. Numbers show the steps where tips are applicable.

<https://doi.org/10.1371/journal.pcbi.1007449.g001>





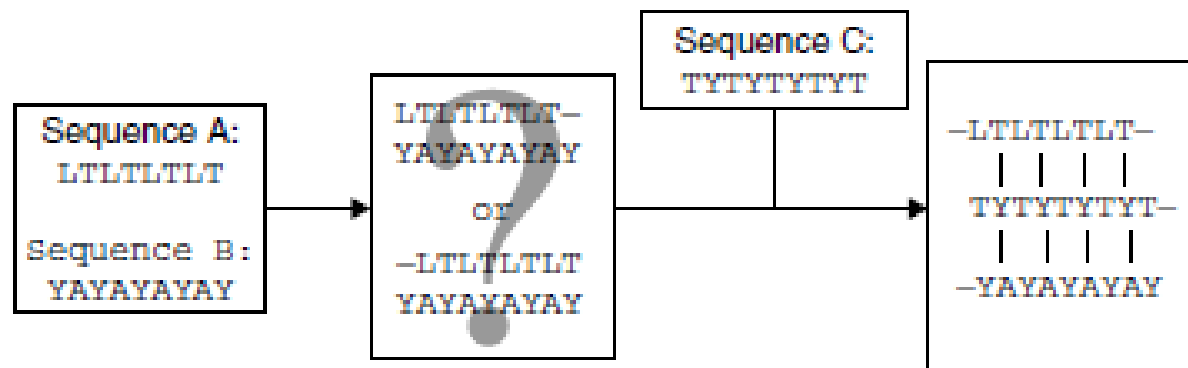


Figure 25.5. A pathological alignment problem. Sequences A and B are impossible to align, unless one considers a third sequence C from a homologous protein.

		1	2	3	4	5	6	7	8	9	10	11	12	13
Template		PHE	ASP	ILE	CYS	ARG	LEU	PRO	GLY	SER	ALA	GLU	ALA	VAL
Model (bad)	1	PHE	ASN	VAL	CYS	ARG	ALA	PRO	---	---	---	GLU	ALA	ILE
Model (good)	2	PHE	ASN	VAL	CYS	ARG	---	---	---	ALA	PRO	GLU	ALA	ILE

Figure 25.6. Example of a sequence alignment where a three-residue deletion must be modeled. While the first alignment appears better when considering just the sequences (a matching proline at position 7), a look at the structure of the template leads to a different conclusion (Figure 25.7).

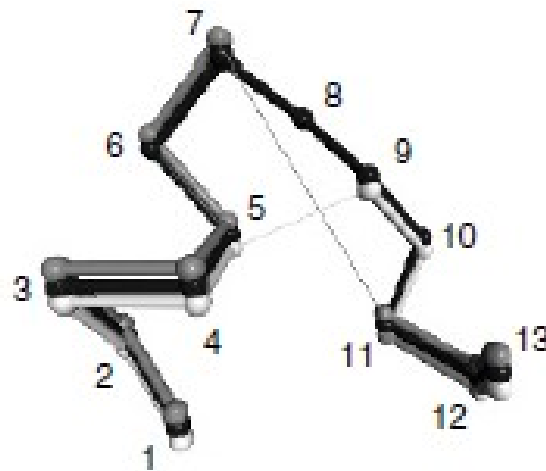
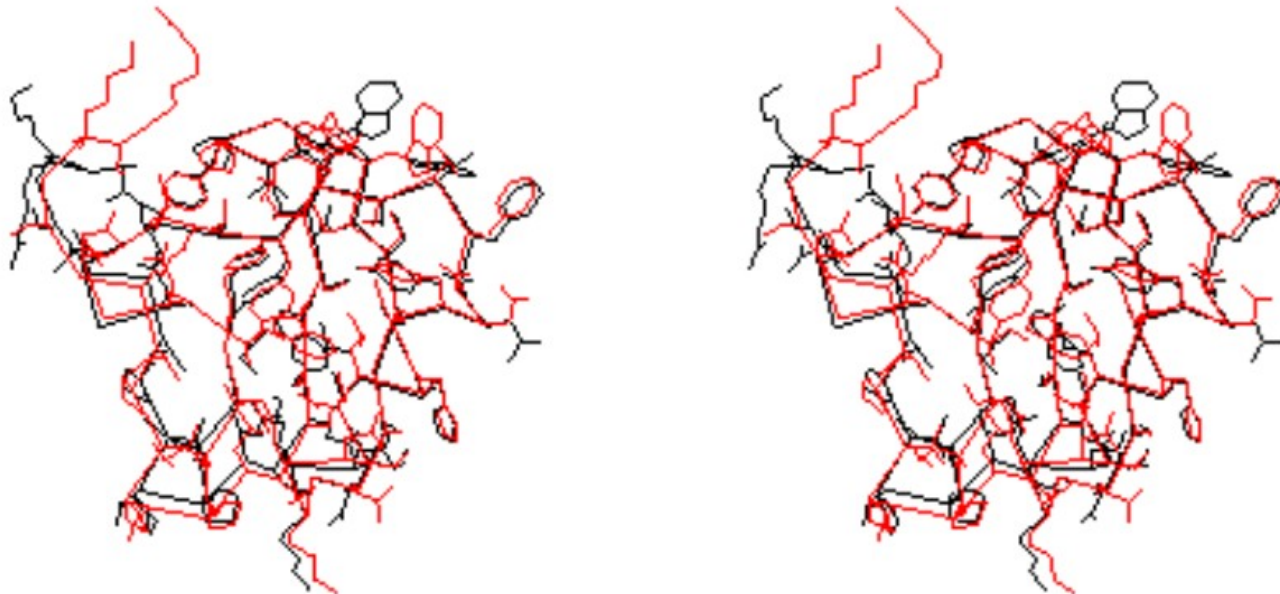


Figure 25.7. Correcting an alignment based on the structure of the modeling template ($C\alpha$ -trace shown in black). While the alignment with the highest score (dark gray, also in Figure 25.6) leads to a gap of 7.5 Å between residues 7 and 11, the second option (white) creates only a tiny hole of 1.3 Å between residues 5 and 9. This can easily be accommodated by small backbone shifts. (The normal $C\alpha-C\alpha$ distance of 3.8 Å has been subtracted).

ACCURACY OF HOMOMOLOGY MODELLING

- ❑ Works for >30% sequence similarity

Example: SWISS-MODEL Prediction of neurotoxin of red scorpion (1DQ7) from neurotoxin of yellow scorpion (1PTX)



DEEP LEARNING METHODS

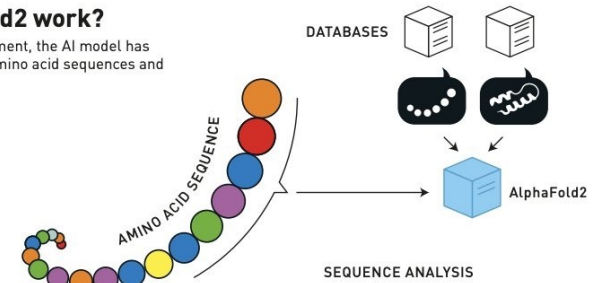
ALPHA FOLD D

How does AlphaFold2 work?

As part of AlphaFold2's development, the AI model has been trained on all the known amino acid sequences and determined protein structures.

1. DATA ENTRY AND DATABASE SEARCHES

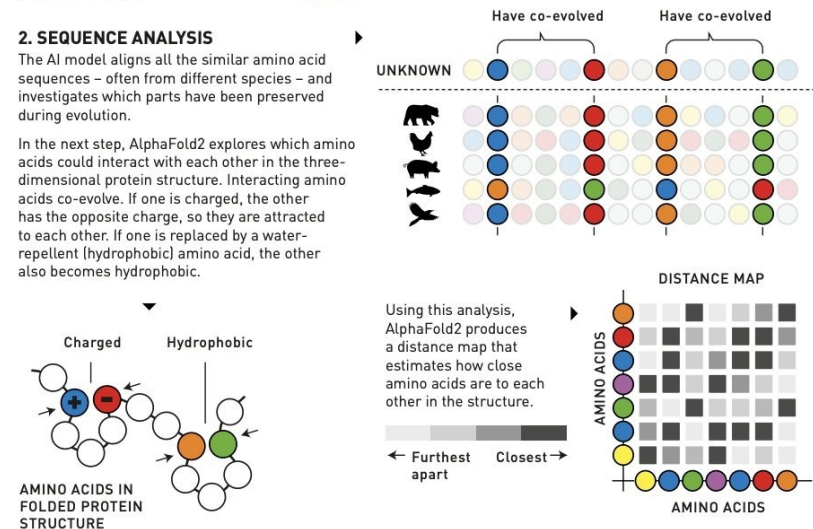
An amino acid sequence with unknown structure is fed into AlphaFold2, which searches databases for similar amino acid sequences and protein structures.



2. SEQUENCE ANALYSIS

The AI model aligns all the similar amino acid sequences – often from different species – and investigates which parts have been preserved during evolution.

In the next step, AlphaFold2 explores which amino acids could interact with each other in the three-dimensional protein structure. Interacting amino acids co-evolve. If one is charged, the other has the opposite charge, so they are attracted to each other. If one is replaced by a water-repellent (hydrophobic) amino acid, the other also becomes hydrophobic.

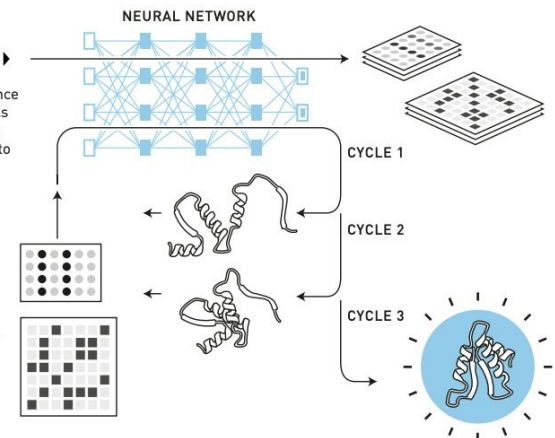


3. AI ANALYSIS

Using an iterative process, AlphaFold2 refines the sequence analysis and distance map. The AI model uses neural networks called transformers, which have a great capacity to identify important elements to focus on. Data about other protein structures – if they were found in step 1 – is also utilised.

4. HYPOTHETICAL STRUCTURE

AlphaFold2 puts together a puzzle of all the amino acids and tests pathways to produce a hypothetical protein structure. This is re-run through step 3. After three cycles, AlphaFold2 arrives at a particular structure. The AI model calculates the probability that different parts of this structure correspond to reality.



The Nobel Prize in Chemistry 2024 was divided, one half awarded to David Baker "for computational protein design", the other half jointly to Demis Hassabis and John Jumper "for protein structure prediction"



© Nobel Prize Outreach. Photo:
Clément Morin

David Baker



© Nobel Prize Outreach. Photo:
Clément Morin

Demis Hassabis



© Nobel Prize Outreach. Photo:
Clément Morin

John Jumper

08/08/2026 Prize share: 1/2

Protein Modeling

Prize share: 1/4

Prize share: 1/4